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Monetary Policy Transmission through the Exchange Rate Factor Structure

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1 Big picture

- **Question.** How is US monetary policy transmitted internationally through currency markets, global intermediaries, and foreign firms' real decisions?
- **Core mechanism.** After an unexpected US monetary easing, global investors and banks rebalance toward riskier currencies. Funds sell safe, low-carry-beta currencies and buy riskier, high-carry-beta currencies; US global banks similarly tilt foreign loan origination toward high-risk currencies.
- **FX factor-structure view.** The paper treats the dollar, carry, and related currency factors as a compact way to summarize systematic currency risk. Currency betas determine whether a currency is “safe” or “risky” in the cross-section.
- **Main data innovation.** The authors combine high-frequency monetary policy shocks, CLS settlement data on currency flows by institution type, syndicated loan origination data from DealScan, and international firm balance-sheet data from Compustat.
- **Main funds result.** Investment funds respond to US monetary easing by purchasing high-carry-beta currencies and selling low-carry-beta currencies. This response is persistent for several months.
- **Main bank-lending result.** Global US banks increase foreign loan origination relatively more in high-carry-beta currencies after US monetary easing. The response is modest on average but meaningful across the currency risk spectrum.
- **Firm real effects.** Firms that primarily operate and borrow in high-risk currencies receive relatively larger loan supply shocks after easing, expand debt and assets, and increase tangible investment.
- **Alternative stories addressed.** The paper checks that the pattern is not just exchange-rate movements, interest-rate spread changes, other currency factors, trend chasing, foreign central bank reactions, or European monetary policy shocks.
- **Economic takeaway.** The risk factor exposure of currencies is an international transmission channel of US monetary policy. US easing does not simply expand global credit uniformly; it reallocates capital across currencies according to systematic FX risk.

2 Incremental contribution of the paper

- **From returns to quantities.** Prior exchange-rate factor papers focus mainly on currency returns and risk premia. This paper shows that the same factor structure organizes *quantities*: actual currency order flows and bank loan origination.
- **A risk-taking channel through currencies.** The paper extends the monetary-policy risk-taking literature by showing that US monetary easing reallocates capital across currencies according to their carry and dollar risk exposures.
- **New link from FX markets to corporate finance.** The paper follows the chain from FOMC shocks to currency flows, from currency flows to US banks' foreign-currency loan origination, and from bank lending to foreign firms' leverage and investment.
- **Institutional granularity.** Using CLS data, the paper separates investment funds, corporates, non-bank financials, non-dealer banks, and dealer banks. This matters because the risk-on response is concentrated among investment funds, while banks act differently across dealer and non-dealer categories.
- **Operating-currency identification.** The firm-level design distinguishes a firm's operating currency from its country of domicile. This allows the paper to compare firms in the same country and quarter but exposed to different currencies.
- **Cross-market unification.** The same carry-beta ranking explains patterns in funds' spot currency trades, global US bank syndicated lending, and firm investment outcomes. This gives a unified explanation for a broad set of international transmission facts.

- **Policy relevance.** The paper suggests that a central bank’s easing can shift global risk allocation toward countries and firms connected to high-risk currencies, rather than simply loosening financial conditions equally everywhere.

3 Data and measurement

3.1 Monetary policy shocks

The monetary policy shock is the surprise component of changes in expected Federal Funds rates around FOMC announcements, following Kuttner (2001). The shock is:

$$MPS_t = (ff0_t - ff0_{t-1}) \frac{m}{m - t},$$

where $ff0_t$ is the current-month Federal Funds futures price and m is the number of days in the month. A positive MPS_t corresponds to a monetary easing, because an increase in the futures price means lower expected rates.

Interpretation: The high-frequency construction isolates unexpected policy news around FOMC meetings. This is important because lower-frequency interest-rate changes could be contaminated by exchange-rate movements, macro news, or foreign capital flows.

3.2 Currency factor exposures

The paper focuses on two main currency risk exposures.

- **Carry beta:** exposure to the carry trade factor, which goes long high-interest-rate currencies and short low-interest-rate currencies.
- **Dollar beta:** exposure to the common dollar factor, the average appreciation or depreciation of currencies against the US dollar.

Currency excess returns are summarized as:

$$\Delta e_{i,t} = \beta'_i f_t + \varepsilon_{i,t},$$

where f_t contains currency risk factors and β_i measures currency i ’s exposure.

Interpretation: High carry-beta currencies are risky in the sense that they load positively on the carry factor and tend to pay higher expected returns. Low or negative carry-beta currencies behave more like safe currencies.

3.3 CLS currency-flow data

The currency-flow data come from Continuous Linked Settlement (CLS), covering September 2012 to March 2024. CLS reports hourly aggregate trading volume and order flow by currency and institution type. The key measure is directional order flow:

$$OF(Y_{i,j,t}) = \text{Buy Volume}_{i,j,t}^{\$} - \text{Sell Volume}_{i,j,t}^{\$},$$

where i is the currency pair and j is the institution type.

The paper separates five institution groups:

- corporates;

- investment funds;
- non-bank financial institutions;
- non-dealer banks;
- dealer banks.

Interpretation: The institution split is central. It lets the paper ask whether monetary policy triggers risk-taking by funds, market-making by dealers, or loan-related rebalancing by banks.

3.4 DealScan syndicated loan data

The bank-lending data come from Thomson Reuters DealScan from January 1999 to March 2024. The paper focuses on syndicated loans that involve at least one global US bank and that are denominated in one of the G10 non-US currencies. Loans are aggregated by bank, currency, and month. The key outcome is the natural log of the dollar amount lent in currency i during month t .

Interpretation: The DealScan data allow the authors to test whether the risk-on pattern observed in currency flows also appears in primary credit markets. This is the bridge from financial flows to real firm outcomes.

3.5 Firm-level balance-sheet data

Firm outcomes are from Compustat North America and Compustat Global from 1999 to 2024. The final firm-level sample contains roughly 50,000 firm-quarter observations and 987 firms that borrow from US banks. Outcomes include:

- borrowing probability;
- debt growth;
- asset growth;
- property, plant, and equipment growth;
- leverage growth;
- Tobin’s Q growth.

Interpretation: The firm-level data allow the paper to move beyond capital-flow correlations. If high-risk-currency firms borrow more from US banks after easing and then expand assets and investment, the FX factor channel has real effects.

4 Models and empirical specifications

4.1 Flow betas: immediate response of currency flows

The first empirical object is the currency-flow beta:

$$OF(S_{i,j,t}) = a_{ij} + \beta_{ij}MPS_t + \varepsilon_{i,j,t}.$$

Here $OF(S_{i,j,t})$ is order flow in currency i by institution type j in month t . The coefficient β_{ij} measures how many dollars flow into the foreign currency after a one-basis-point easing.

Interpretation: A positive β_{ij} means the institution buys the foreign currency and sells dollars when US monetary policy eases. A negative beta means it sells that foreign currency and buys dollars.

4.2 FX factor interaction for funds

To explain cross-currency heterogeneity in funds' flows, the paper estimates:

$$OF(S_{i,t}) = \mu_i + \alpha_t + \gamma X_{i,t} + \beta MPS_t + \varphi(X_{i,t} \times MPS_t) + \kappa W_{i,t} + \varepsilon_{i,t}.$$

Here $X_{i,t}$ is a currency risk characteristic such as carry beta, dollar beta, or UMVE beta; $W_{i,t}$ includes controls such as bid-ask spreads and exchange-rate changes.

Interpretation: The interaction coefficient φ is the main object. A positive carry-beta interaction means investment funds buy more of high-carry-beta currencies after monetary easing.

4.3 Persistence of currency-flow effects

The paper estimates local projections:

$$OF(S_{i,t+h}) = \alpha_i^h + \sum_{m=0}^3 \beta_m^h MPS_{t-m} + \varepsilon_{i,t+h},$$

where h indexes months after the policy shock.

Interpretation: This specification asks whether the risk-on response is a one-day trading blip or a persistent reallocation. The paper finds persistence over several months.

4.4 Bank loan origination and currency risk

The bank-lending analogue regresses loan origination on monetary policy shocks and currency characteristics:

$$\log Loan_{i,t} = \mu_i + \alpha_t + \gamma X_{i,t} + \beta MPS_t + \varphi(X_{i,t} \times MPS_t) + \psi \Delta \log S_{i,t} + \varepsilon_{i,t}.$$

Interpretation: The key question is whether banks tilt loan origination toward high-risk currencies after US easing. The positive carry-beta interaction implies they do.

4.5 Firm-level borrowing and real outcomes

The firm-level borrowing specification is:

$$D_{j,i,c,t} = \mu_j + \alpha_{c,t} + \nu_{n,t} + \gamma X_{i,t} + \varphi(X_{i,t} \times MPS_t) + \psi \Delta \log S_{i,t} + \kappa W_{j,t} + \varepsilon_{j,i,c,t},$$

where $D_{j,i,c,t}$ is a borrowing indicator for firm j in country c borrowing in currency i .

For real outcomes, the dependent variables are changes in debt, assets, and PPE:

$$\Delta \log Y_{j,i,c,t} = \mu_j + \alpha_{c,t} + \nu_{n,t} + \gamma X_{i,t} + \varphi(X_{i,t} \times MPS_t) + \psi \Delta \log S_{i,t} + \kappa W_{j,t} + \varepsilon_{j,i,c,t}.$$

Interpretation: Country-quarter fixed effects absorb local demand and macro conditions; industry-quarter fixed effects absorb sectoral shocks. Identification comes from firms in the same country and quarter that operate in different currencies with different carry betas.

5 Identification and empirical design

5.1 What identifies the currency-flow response

The currency-flow response is identified from high-frequency FOMC surprise shocks interacted with predetermined currency risk exposures. The key assumption is that within the narrow FOMC window, unexpected changes in Federal Funds futures are not driven by contemporaneous currency flows.

Interpretation: This is the standard high-frequency monetary policy identification logic. It isolates exogenous policy news from endogenous policy responses to macroeconomic conditions.

5.2 What identifies the bank-lending response

The bank-lending response is identified from cross-currency differences in factor exposures. The paper asks whether, after the same US monetary shock, global US banks increase lending relatively more in high-carry-beta currencies than in low-carry-beta currencies.

Interpretation: This is not simply a before-after effect of easing. It is a cross-sectional slope: the same shock has different lending effects depending on the currency's risk beta.

5.3 What identifies firm real effects

The firm analysis exploits the fact that firms outside the US can borrow and operate in currencies that differ from their country of domicile. Country-quarter fixed effects absorb country-level demand shocks and local macroeconomic conditions. Industry-quarter fixed effects absorb sector-specific fluctuations.

Interpretation: The comparison is therefore between firms that share the same country and quarter but differ in operating currency exposure. This helps isolate the loan-supply channel from local demand.

5.4 Threats and how the paper addresses them

- **Exchange-rate valuation effects.** The regressions control for changes in spot exchange rates so that flows are not mechanically driven by currency appreciation or depreciation.
- **Interest-rate spread effects.** The paper shows that the results are not simply due to lower interest-rate spreads in high-risk currencies.
- **Other currency-risk factors.** The authors test other factors such as momentum, value, output gaps, skewness, and UMVE betas.
- **Foreign central bank responses.** Table 13 checks whether foreign central banks respond to Fed policy in a way that explains the results.
- **European monetary policy.** Table 14 shows that euro-based currency pairs do not exhibit the same systematic response to ECB shocks.
- **Persistence of exposures.** Online appendix checks show that carry and dollar betas are persistent, supporting their interpretation as pre-existing currency risk exposures.

Interpretation: The identification strategy is strongest when the same sign pattern appears across three layers: investment funds' currency flows, banks' loan origination, and firm-level leverage and investment.

6 Tables and figures

6.1 Table 1: Summary statistics of monetary policy shocks; Fig. 1: Snapshot of CLS data; Table 2: Summary statistics – CLS

Read: Table 1 summarizes monetary policy shocks at high-frequency, monthly, and quarterly frequencies. Fig. 1 shows how CLS classifies transactions and computes order flow. Table 2 shows that investment funds and non-dealer banks generate substantial variation in currency order flow.

Detailed interpretation: The high-frequency shocks have a mean close to zero and a large standard deviation, which is useful for event-study identification. The CLS snapshot clarifies that order flow is measured as buy volume minus sell volume in dollar terms. Table 2 shows that funds have the largest standard deviations among price-taking institutions, making them the natural place to look for risk-on trading.

How it fits the paper: These visuals set up the first empirical step: measure exogenous US policy shocks and measure which institutions buy or sell currencies after those shocks.

Table 1
Summary statistics of monetary policy shocks.

	High-frequency	Monthly		Quarterly	
		Simple sum	Weighted sum	Simple sum	Weighted sum
Mean	1.18	1.20	0.81	2.43	1.20
Median	0.00	0.00	0.00	0.00	0.00
Std.	8.30	9.03	7.79	11.76	8.97
Min.	-24.89	-24.89	-24.89	-29.19	-23.50
Max.	74.06	83.24	72.43	66.55	51.47
#Obs	208	204	204	101	101

Note: This table reports summary statistics of monetary policy shocks following [Kuttner \(2001\)](#). All numbers are in basis points (bps), except for the row labeled “#Obs”, which shows the number of monetary policy shocks over the sample period from 1 January 1999 to 31 March 2024. The first column labeled “High-frequency” refers to shocks that are estimated using the event study strategy in Eq. (3). The columns labeled “Simple sum” aggregate the high-frequency shocks by simply summing up all shocks within a month or quarter, whereas the columns labeled “Weighted sum” aggregate the shocks to a monthly or quarterly frequency using the *unweighted means method* described in the main text.

Table 1: monetary policy shock summary.

Table 2
Summary statistics — CLS.

	Corporates		Funds		NBFIs		Non-dealer banks	
	Std.	Share	Std.	Share	Std.	Share	Std.	Share
USDAUD	0.42	0.36	2.65	10.98	0.46	3.18	3.97	85.49
USDCAD	0.70	0.29	15.58	10.40	1.03	1.98	31.40	87.33
USDCHE	0.57	0.90	2.41	9.06	1.45	4.17	4.27	85.87
USDEUR	3.46	2.19	11.39	13.78	1.45	3.18	14.67	80.85
USDGBP	1.23	1.00	5.82	13.02	1.56	3.56	7.96	82.42
USDJPY	0.94	0.85	4.81	8.93	0.96	3.14	6.19	87.08
USDNOK	0.12	0.45	0.58	12.75	0.10	2.94	1.55	83.86
USDNZD	0.04	0.08	1.18	7.30	0.15	3.46	1.68	89.16
USDSEK	0.18	1.18	0.97	20.78	0.12	2.78	1.64	75.26

Note: This table collects summary statistics for the CLS order flow data. The columns labeled *Std.* report the standard deviation of monthly order flows (buy volume minus sell volume) in \$bn broken down by four categories of market participants, namely, corporates, funds, non-bank financials (NBFIs), and non-dealer banks. The columns labeled *Share* are computed based on the sum of buy and sell volume and reflect the relative share (summing up to 100% for each currency pair) in percent of trading volume associated with each of the four groups of market participants. The sample covers the period from September 2012 to March 2024.

Table 2: CLS order-flow summary by institution.

Data construction and institutional flow variation.

6.2 Fig. 2: Snapshot of DealScan data; Table 3: Summary statistics – Compustat

Read: Fig. 2 shows that US banks’ foreign lending is concentrated in a handful of currencies, especially EUR and GBP. Table 3 summarizes firm-quarter outcomes and borrowing variables for the final Compustat sample.

Detailed interpretation: The DealScan figure shows that loan origination is not evenly distributed across currencies. This matters because the paper’s cross-currency comparisons require enough activity in several currencies. Table 3 shows that the firm sample is sizeable and contains meaningful variation in debt growth, asset growth, PPE growth, and loan amounts.

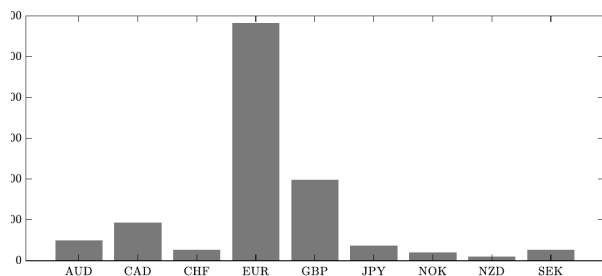
Instrument	London Date	Hour	Price Taker	Market Maker	Buy CCY	Sell CCY	Buy Volume	Sell Volume	Net Volume	Order Flow
SPT	2019-01-02	12	BuySide	SellSide	EUR	USD	1,097,027,686	2,245,621,140	-647,603,463	-732,701,707
SPT	2019-01-02	12	Corporate	Bank	EUR	USD	86,656,909	9,786,133	-78,873,666	-89,221,229
SPT	2019-01-02	12	Paul	Bank	EUR	USD	42,308,060	634,132,478	-591,824,418	-668,628,535
SPT	2019-01-02	12	Non-Bank Financial	Bank	EUR	USD	31,007,366	132,101,121	-101,093,755	-81,445,554
SPT	2019-01-02	12	Non-Dealer Bank	Bank	EUR	USD	1,415,095,242	1,460,599,817	-54,504,575	-61,658,846

S data.

denote the “residual” calculated to define non-dealer banks. For buy volume, it is calculated by subtracting the volume from total buy-side (price taker) activity; there is a similar calculation for sell volume. The EURUSD exchange is EUR.

Fig. 1: CLS snapshot and order-flow construction.

How it fits the paper: These visuals connect the currency-flow analysis to the bank-loan and firm-balance-sheet analyses.



DealScan data.
resents the average of the total aggregate loan amount in a given currency aggregated over a month. The sample covers the period from January 1999 to March 2024.

Table 3

Fig. 2: DealScan loan amounts by currency.

Loan and firm-level samples.

Table 3

Summary statistics — Compustat.

	#Obs	Mean	Std.	5%	25%	50%	75%	95%
Current asset ratio in %	50,494	34.4	19.6	5.9	19.1	32.5	47.9	68.8
Book leverage in %	50,498	30.0	21.0	4.7	18.7	28.0	39.0	59.2
$\Delta \log \text{ debt in \%}$	50,498	1.8	34.3	-23.2	-4.5	0.0	4.8	32.4
$\Delta \log \text{ assets in \%}$	50,498	1.7	13.8	-10.1	-2.5	0.9	4.4	15.2
$\Delta \log \text{ PPE in \%}$	50,275	1.6	24.2	-11.3	-2.5	0.6	4.2	16.8
$\Delta \log \text{ sales in \%}$	50,498	1.4	25.3	-28.1	-4.7	1.1	7.7	31.3
$\Delta \log \text{ leverage in \%}$	50,498	0.0	4.4	-4.5	-1.0	0.0	0.9	4.6
$\Delta \log \text{ Tobin's Q in \%}$	50,498	-0.2	20.6	-20.8	-4.8	0.3	5.2	19.2
Loan amount in \$mn	2,116	382.9	686.8	26.8	72.3	159.7	391.2	1,402.2
$\log (\text{loan amount})$	2,116	19.0	1.2	17.1	18.1	18.9	19.8	21.1
Borrowing probability in %	N/A	3.1	3.4	0.0	1.0	2.2	4.3	9.2

Note: This table reports summary statistics for our sample of firms borrowing from US banks. It compiles the average *Mean*, standard deviation *Std.*, and the 5, 25, 50, 75 and 95 percentile of several firm-level variables: *current asset ratio* is the ratio of current to total assets, *book leverage* is book debt over total assets, $\Delta \log \text{ debt}$ is the relative change in firm debt (short-term plus long-term liabilities), $\Delta \log \text{ assets}$ is the relative change in total assets, $\Delta \log \text{ PPE}$ is the relative change in property plant and equipment (PPE) net of depreciation, $\Delta \log \text{ sales}$ is the relative change in total gross sales net of trade and cash discounts, $\Delta \log \text{ leverage}$ is the relative change in leverage, $\Delta \log \text{ Tobin's Q}$ is the relative change in Tobin's Q ((market value of equity + book value of debt)/book value of total assets), *loan amount* is the aggregate amount of the portion of all syndicated loans that a firm has borrowed in a given quarter from US banks, $\log \text{ loan amount}$ is the log of loan amount, and *borrowing probability* is the share of quarters during which a firm borrows from US banks in the same currency as the reporting currency of the firm. #Obs counts the number of non-missing observations for each variable, except for loan amount where it counts non-zero observations. The sample covers the period from January 1999 to March 2024.

Table 3: Compustat firm summary.

6.3 Table 4: Flow betas for G10 currency pairs; Fig. 3: Funds' currency flow betas and carry betas; Table 5: Currency flows of funds and the FX factor structure

Read: Table 4 shows the immediate currency-flow response by institution type. Fig. 3 plots funds' flow betas against carry betas. Table 5 estimates panel regressions showing that fund flows respond more strongly to monetary easing in high-carry-beta currencies.

Detailed interpretation: The most important pattern in Table 4 is that funds have positive flow betas for high-carry currencies such as AUD and CAD and negative flow betas for safe currencies such as EUR. Fig. 3 makes this cross-sectional slope visually clear. Table 5 confirms the pattern after controlling for currency and time fixed effects, bid-ask spreads, and exchange-rate changes.

How it fits the paper: This is the paper's first core mechanism: monetary easing induces a risk-on currency reallocation by investment funds.

	Corporates	Funds	NBIFs	Non-dealer banks	Dealer banks	Carry beta	Dollar beta	FX beta
USDJPY	8.04 [0.56]	35.88 [0.74]	14.42 [1.03]	-246.79*** [4.73]	188.45*** [4.25]	-0.26 [0.08]	0.51 [0.08]	-0.13 [0.08]
USDCNH	7.08 [0.76]	23.51 [0.42]	-10.85 [0.67]	-15.52 [0.57]	-4.22 [0.10]	-0.10 [0.08]	0.98 [0.08]	-0.15 [0.08]
USDEUR	31.47 [0.95]	-580.18*** [4.57]	16.15* [1.86]	-62.86 [0.40]	595.42** [2.44]	-0.09 [0.04]	1.11 [0.07]	-0.25 [0.03]
USDGBP	-72.82*** [2.86]	210.39*** [3.38]	-25.46* [1.76]	27.04 [0.18]	-139.15 [1.29]	0.04 [0.05]	0.87 [0.05]	0.03 [0.05]
USDSEK	4.17*** [2.54]	-30.21*** [3.14]	-2.18 [1.38]	20.67 [1.03]	7.55 [0.51]	0.05 [0.05]	1.34 [0.05]	-0.15 [0.05]
USDNOK	-4.39*** [2.12]	-13.99*** [3.33]	-1.19 [1.15]	-47.20*** [4.60]	66.84*** [6.47]	0.21 [0.04]	1.57 [0.06]	0.15 [0.06]
USDCAD	2.18 [0.46]	537.22*** [5.67]	25.06 [1.51]	500.92 [1.62]	-1,065.39*** [13.30]	0.28 [0.08]	0.96 [0.08]	-0.30 [0.08]
USDNZD	-1.67 [1.59]	23.94 [1.13]	1.32 [1.55]	-49.58** [2.11]	26.00*** [2.96]	0.51 [0.08]	1.42 [0.08]	0.50 [0.08]
USDAUD	-0.02 [0.00]	190.42*** [5.23]	1.03 [0.20]	-213.38*** [3.42]	21.95 [0.64]	0.58 [0.08]	1.44 [0.08]	0.37 [0.08]

Note: This table reports the β_j regression coefficients from $OF(S_{i,t}) = \alpha_i + \beta_j MPS_t + \epsilon_{i,t}$, where $OF(S_{i,t})$ is the currency flow in \$bn in currency pair i / customer group j in month t and MPS_t is our monetary policy shock measure in basis points that we extract from Fed Fund futures rate changes following [Kuttner \(2001\)](#). Currency pairs are sorted by carry betas in ascending order. By market clearing, the column labeled "dealer banks" is equal to the sum of the first four columns times minus one. The penultimate two columns report the average carry and dollar betas that we compute based on rolling window regressions. The last column reports the exchange rate response to a standard deviation easing surprise of MPS_t ([Dooley et al., 2017](#)); positive estimates indicate an appreciation of the foreign currency. Asterisks *, **, and *** denote significance at the 10%, 5%, and 99% confidence.

Table 4: flow betas by currency and institution.

Table 5

Currency flows of funds and the FX factor structure.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
carry $\beta_{i,t}$		0.02 [0.39]			-0.01 [0.05]		-0.15 [1.10]	
dollar $\beta_{i,t}$			0.13 [0.70]			0.14 [0.76]	0.20 [0.97]	
UMVE $\beta_{i,t}$				-0.19 [1.55]				-0.13 [1.02]
MPS_t	0.03 [0.86]	0.01 [0.26]	-0.04 [0.43]	0.08 [1.61]				
carry $\beta_{i,t} \times MPS_t$		0.09*** [2.73]			0.09*** [2.61]		0.17*** [4.18]	
dollar $\beta_{i,t} \times MPS_t$			0.08 [1.13]			0.07 [0.95]	-0.36*** [3.68]	
UMVE $\beta_{i,t} \times MPS_t$				0.08*** [3.76]				0.08*** [4.92]
$\Delta \log \text{bid-ask spread}_{i,t}$	-0.03 [1.34]	-0.03 [1.57]	-0.03 [1.52]	-0.03 [1.52]	-0.01 [0.67]	-0.02 [0.87]	0.03 [0.92]	-0.01 [0.95]
$\Delta \log S_{i,t}$	-0.01 [0.38]	-0.02 [0.52]	-0.03 [0.97]	-0.03 [0.97]	-0.02 [0.44]	-0.03 [0.59]	-0.02 [0.47]	-0.03 [0.74]
Overall R^2 in %	18.76	19.45	19.25	21.33	30.80	30.64	31.88	31.24
Total # of Obs.	1251	1242	1242	1242	1242	1242	1242	1242
Avg. #Time periods	139	138	138	138	138	138	138	138
#Currencies	9	9	9	9	9	9	9	9
Currency FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time series FE	No	No	No	No	Yes	Yes	Yes	Yes

Note: This table reports results from fixed effects panel regressions of the form $OF(S_{i,t}) = \mu_i + \alpha_i + \gamma X_{i,t} + \beta MPS_t + \varphi(X_{i,t} \times MPS_t) + \kappa W_{i,t} + \epsilon_{i,t}$, where $OF(S_{i,t})$ is the order flow by funds in \$bn in currency pair i in month t . $X_{i,t}$ denotes either the carry $\beta_{i,t}$, dollar $\beta_{i,t}$, or UMVE $\beta_{i,t}$ that are based on rolling window regressions of currency returns on the carry, dollar, and UMVE factor, respectively. MPS_t is our monetary policy shock in basis points that we extract from Fed Fund futures rate changes following [Kuttner \(2001\)](#). $W_{i,t}$ may include the following control variables: $\Delta \log \text{bid-ask spread}_{i,t}$ is the log change in the monthly average relative bid-ask spread and $\Delta \log S_{i,t}$ is the log change in the spot exchange rate expressed as the number of foreign currency units per unit of US dollar. Both dependent and independent variables are measured in units of standard deviations. The test statistics based on double clustered (by currencies and time) standard errors, allowing for serial correlation up to 3 lags are reported in brackets. Asterisks *, **, and *** denote significance at the 90%, 95%, and 99% confidence levels. The sample covers the period from September 2012 to March 2024.

Table 5: fund-flow regressions with currency risk factors.

Fund flows and the FX factor structure.

6.4 Fig. 4: Persistent effect of monetary policy on spot currency flows; Fig. 5: Foreign credit betas and carry betas; Table 6: International lending and the FX factor structure

Read: Fig. 4 shows that low-carry currencies experience persistent outflows after easing while high-carry currencies experience persistent inflows. Fig. 5 shows that bank credit betas also rise with carry betas. Table 6 estimates this relationship in panel regressions.

Detailed interpretation: Fig. 4 is important because the response lasts beyond the announcement month, making it relevant for financing conditions. Fig. 5 shows that banks' credit allocation tilts in the same direction as funds' currency flows, although the relationship is less visually sharp. Table 6 confirms that the interaction of carry beta with the monetary policy shock is positive and significant in several specifications.

How it fits the paper: This section moves the mechanism from secondary-market currency trading to primary-market credit origination.

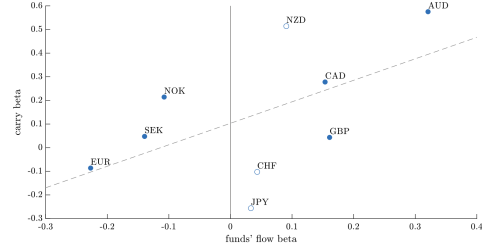


Fig. 3: Funds' currency flow betas and carry betas.
Note: This figure plots the β_j regression coefficient estimates from $OF(S_{i,t}) = \alpha_i + \beta_j MPS_t + \epsilon_{i,t}$, where $OF(S_{i,t})$ is the currency flow in \$bn in currency pair i by investment funds in month t and MPS_t is our monetary policy shock measure in basis points against the average carry beta. For the regression, both dependent and independent variables are measured in units of standard deviations. Filled dots indicate point estimates that are statistically significant at the 10% confidence level. The inference is based on robust standard errors ([Newey and West, 1987](#)), correcting for heteroskedasticity and serial correlation up to 3 lags. The sample spans from September 2012 to March 2024.

Fig. 3: funds' flow betas versus carry betas.

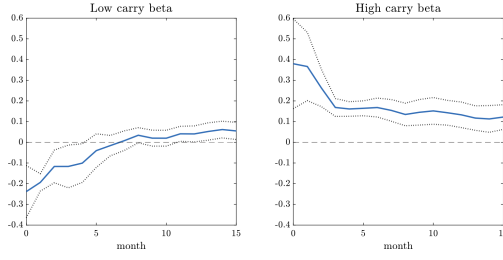


Fig. 4: Persistent effect of monetary policy on spot currency flows.
Note: This figure plots the impulse response function of currency flows to monetary surprises with local projections (Jorda, 2005) from $OF(S_{i,t+h}^c) = \alpha_0^c + \sum_{j=1}^h \rho_j^c MPS_{i,t+j} + \epsilon_{i,t+h}^c$, where $OF(S_{i,t+h}^c)$ is the cumulative order flow in currency group i in \$bn after h months. Both dependent and independent variables are in standardized units. Currency groups are formed based on country i 's carry beta and MPS_i is our monetary policy shock. The dotted lines mark the 90% confidence bands based on Newey and West (1987) standard errors, correcting for the autocorrelation induced by overlapping periods. The sample covers the period from September 2012 to March 2024.

Fig. 4: local-projection impulse responses for flows.

Table 6

International lending and the FX factor structure.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
carry beta _{i,t}		-0.24 [0.82]			-0.80** [1.99]		-1.53* [1.94]	
dollar beta _{i,t}			0.62* [1.70]			0.58 [1.53]	0.97* [1.67]	
UMVE beta _{i,t}				0.30 [1.13]				0.01 [0.03]
MPS _t	0.02 [0.40]	0.05 [1.35]	0.57*** [3.44]	0.05 [1.51]				
carry beta _{i,t} × MPS _t		0.14*** [4.86]			0.15** [1.97]		0.21** [2.36]	
dollar beta _{i,t} × MPS _t			-0.56** [2.33]			-0.67** [2.51]	-0.87*** [2.82]	
UMVE beta _{i,t} × MPS _t				0.15*** [2.87]				0.19** [2.36]
$\Delta \log S_{i,t}$	-0.06 [0.52]	-0.05 [0.47]	-0.07 [0.63]	-0.06 [0.51]	0.00 [0.02]	-0.03 [0.14]	-0.02 [0.11]	0.01 [0.04]
Overall R ² in %	57.16	57.21	57.36	57.26	62.36	62.42	62.73	62.26
Total # of Obs.	2727	2727	2727	2727	2727	2727	2727	2727
Avg. #Time periods	303	303	303	303	303	303	303	303
#Currencies	9	9	9	9	9	9	9	9
Currency FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time series FE	No	No	No	No	Yes	Yes	Yes	Yes

Note: This table reports results from fixed effects panel regressions of the form $\log Loan_{i,t} = \mu_i + \alpha_i + \gamma X_{i,t} + \beta MPS_t + \varphi(X_{i,t} \times MPS_t) + \psi \Delta \log S_{i,t} + \epsilon_{i,t}$, where $\log Loan_{i,t}$ is the natural log of the dollar amount lent by banks domiciled in the US (either bank holding company or subsidiary) to firms domiciled abroad in currency i during month t . We winsorize the dependent variable at the 1% level. $X_{i,t}$ denotes either the carry beta_{i,t}, dollar beta_{i,t}, or UMVE beta_{i,t} that are based on rolling window regressions of currency returns on the carry, dollar, and UMVE factor, respectively. MPS_t is our monetary policy shock in basis points that we extract from Fed Fund futures rate changes following Kuttner (2001). $\Delta \log S_{i,t}$ is the log change in the spot FX rate expressed as the number of foreign currency units per US dollar. The independent variables are measured in units of standard deviations.

Table 6: bank lending and currency factors.

Persistent flows and bank credit reallocation.

6.5 Table 7: Firm-level foreign currency borrowing and the FX factor structure; Table 8: Firm-level outcomes and the FX factor structure

Read: Table 7 shows that firms using high-carry-beta currencies are more likely to borrow after US monetary easing. Table 8 shows that these firms expand debt, assets, and PPE more than low-carry-beta firms.

Detailed interpretation: Table 7 uses firm fixed effects and several country/industry time controls. The average marginal effects imply that exposure to a high-carry currency raises borrowing probability after easing. Table 8 shows that the loan supply shock has balance-sheet consequences: high-carry-beta firms lever up and invest in tangible capital.

How it fits the paper: These are the paper's main real-effects tables. They establish that the FX factor channel does not stop at financial flows; it reaches firm leverage and investment.

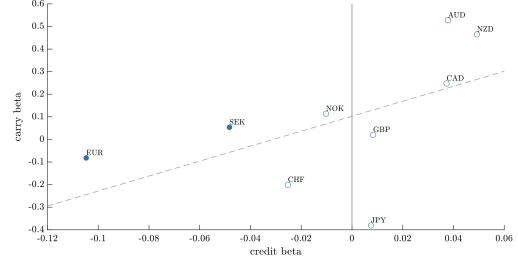


Fig. 5: Foreign credit betas and carry betas.
Note: This figure plots the β regression coefficient estimates from $\log Loan_{i,t} = \alpha_i + \beta MPS_t + \gamma \Delta \log S_{i,t} + \epsilon_{i,t}$ against the average carry beta. $\log Loan_{i,t}$ is the natural log of the dollar amount lent by global banks headquartered in the US to corporations domiciled abroad in foreign currency i during month t . MPS_t is our monetary policy shock measure in basis points (Kuttner, 2001). $\Delta \log S_{i,t}$ is the log change in the spot exchange rate expressed as the number of foreign currency units per unit of US dollar. For the regression, both dependent and independent variables are measured in units of standard deviations. Filled dots indicate point estimates that are statistically significant at the 10% confidence level. The inference is based on robust standard errors (Newey and West, 1987), correcting for heteroskedasticity and first-order serial correlation. The sample spans from January 1999 to March 2024.

Fig. 5: credit betas versus carry betas.

Table 7
Firm-level foreign currency borrowing and the FX factor structure.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
carry beta _{it}	-0.23* [1.83]		-0.06 [0.44]	-0.91* [1.87]		-0.84 [1.52]	-1.14* [1.85]		-1.10* [1.68]
dollar beta _{it}		-0.31*** [4.04]	-0.31*** [1.72]		-0.47 [1.05]	-0.30 [0.62]		-0.42 [0.83]	-0.24 [0.46]
carry beta _{it} × MPS _{it}	0.09*** [2.73]		0.09*** [2.17]	0.71*** [3.84]		0.92*** [2.97]	0.99*** [4.00]		1.19*** [3.10]
dollar beta _{it} × MPS _{it}		-0.30*** [3.77]	-0.24*** [1.60]		0.01 [0.03]	1.67 [1.37]		0.12 [0.05]	1.70 [1.25]
Δ log S _{it}	0.07 [0.26]	0.09 [0.34]	0.04 [0.14]	0.02 [0.03]	-0.02 [0.02]	0.12 [0.13]	-0.23 [0.24]	-0.20 [0.19]	-0.16 [0.19]
log assets _{it,t-1}	0.61*** [4.08]	0.67*** [4.49]	0.67*** [4.51]	0.79*** [4.99]	0.78*** [4.95]	0.79*** [4.97]	0.79*** [4.50]	0.79*** [4.49]	0.79*** [4.53]
current asset ratio _{it,t-1}	0.02 [0.18]	0.06 [0.59]	0.05 [0.55]	0.07 [0.70]	0.07 [0.72]	0.05 [0.71]	0.06 [0.52]	0.06 [0.55]	0.06 [0.43]
Δ log sales _{it}	0.08*** [2.88]	0.08*** [2.80]	0.08*** [2.83]	0.10*** [3.05]	0.10*** [3.07]	0.10*** [3.05]	0.11*** [2.94]	0.11*** [2.96]	0.10*** [2.94]
Δ log leverage _{it}	0.11** [2.32]	0.13** [2.29]	0.13** [2.30]	0.14** [2.31]	0.14** [2.35]	0.14** [2.31]	0.14** [2.36]	0.14** [2.38]	0.14** [2.36]
Δ log Tobin's Q _{it}	-0.10** [2.14]	-0.10** [2.04]	-0.10** [2.05]	-0.10** [2.16]	-0.09** [2.18]	-0.09** [2.16]	-0.17** [2.50]	-0.17** [2.52]	-0.17** [2.50]
AME in %	0.36	-1.11	0.35	2.91	0.04	3.77	4.50	0.53	5.40
Odds-Ratio	1.10	0.74	1.10	2.04	1.01	2.52	2.70	1.12	3.30
Total # of Obs.	50498	50498	50498	50498	50498	50498	50498	50498	50498
Avg. #Time periods	51	51	51	51	51	51	51	51	51
#Firms	987	987	987	987	987	987	987	987	987
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	No	No	No	No	No	No
Country × quarter FE	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Industry × quarter FE	No	No	No	No	No	No	Yes	Yes	Yes

Note: This table reports results from panel logit regressions with fixed effects of the form $D_{it,j} = \mu_i + \alpha_i + \gamma_i X_{it} + \phi(X_{it} \times MPS_{it}) + \psi \Delta \log S_{it} + \omega W_{it} + \epsilon_{it,j}$, where the dependent variable $D_{it,j}$ is a dummy variable that is equal to unity if firm i in country c has borrowed via a syndicate, which involves at least one bank that is domiciled in the US (either bank holding company or subsidiary) in quarter t and currency i is equal to the reporting currency of the firm. As before, X_{it} denotes either the carry beta, or dollar beta, and we also attribute these dollar and carry betas to each firm based on the reporting currency of the firm. MPS_{it} is our monetary policy shock in basis points following [Kuttner \(2001\)](#). $\Delta \log S_{it}$ is the change in the log of the spot FX rate and W_{it} includes the same controls as in Equation 11 in the paper. μ_i , α_i , and γ_i denote firm, quarter times firm, and quarter times industry α fixed effects, respectively. AME in % is the average marginal effect for the point estimate concerning either the interaction of dollar or carry beta and $\Delta \log S_{it}$. Odds-Ratio is computed as the exponential of the estimated interaction effects. Columns 7-9, and 4 report the AME in %, and Odds-Ratio.

Table 7: borrowing probability and currency risk.

Firm-level borrowing and real outcomes.

6.6 Table 9: Currency flows of funds and economic sources of currency risk; Table 10: Currency flows of funds and a comparison of currency risks; Fig. 6: Currency flow betas and carry betas – monetary easing and tightening

Read: Table 9 compares carry beta to deeper economic risk measures such as global imbalances, size, centrality, and import ratios. Table 10 compares carry with other currency risk factors such as momentum, value, output gaps, and skewness. Fig. 6 separates monetary easing and tightening.

Detailed interpretation: The carry factor remains the most robust cross-sectional summary for the risk-on response. Alternative factors sometimes matter, but they do not explain away the carry-beta pattern. Fig. 6 shows asymmetry: the risk-on relation is stronger during monetary easing than tightening.

How it fits the paper: These robustness checks strengthen the interpretation that the results are about systematic currency risk rather than generic trend chasing or one specific factor construction.

Table 9
Currency flows of funds and economic sources of currency risk.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
MPS _{it}	0.01 [0.26]	0.01 [0.53]	0.04 [1.12]	0.10** [2.07]	-0.03 [0.46]	0.09** [2.06]	0.12*** [2.83]	-0.0 [0.11]
carry beta _{it} × MPS _{it}	0.09*** [2.73]							
IMB beta _{it} × MPS _{it}		-0.01 [0.48]						
downside beta _{it} × MPS _{it}			0.01 [0.72]					
$f_{it} - s_{it} \times MPS_{it}$				0.09** [2.01]				
centrality _{it} × MPS _{it}					-0.08 [1.40]			
term premium _{it} × MPS _{it}						0.09 [1.60]		
size _{it} × MPS _{it}							-0.14*** [12.90]	0.0
import ratio _{it} × MPS _{it}								0.0 [4.9]
Δ log bid-ask spread _{it}	-0.03 [1.34]	0.00 [0.52]	-0.03 [1.47]	-0.03 [1.56]	-0.04 [1.62]	-0.03 [1.62]	-0.04* [1.69]	-0.0 [1.0]
Δ log S _{it}	-0.01 [0.38]	-0.03 [0.72]	-0.02 [0.44]	-0.02 [0.52]	-0.02 [0.50]	-0.02 [0.49]	-0.01 [0.39]	-0.0 [0.6]
Overall R ² in %	19.45	34.52	19.19	19.22	19.73	19.34	21.69	33.5
Total # of Obs.	1242	846	1242	1242	1242	1242	1215	891
Avg. #Time periods	9	9	138	138	138	138	135	99
#Currencies	9	9	9	9	9	9	8	9
Currency FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note: This table reports results from fixed effects panel regressions of the form $OF(S_{it}) = \mu_i + \gamma_i X_{it} + \beta MPS_{it} + \phi(X_{it} \times MPS_{it}) + \alpha W_{it} + \epsilon_{it}$, where $OF(S_{it})$ is the β flow by funds in \$ in currency pair i in month t . X_{it} denotes various measures of country level risk characteristics. For convenience, we only report the β and ϕ coefficients. MPS_{it} is our monetary policy shock in basis points that we extract from Fed Fund futures rate changes following [Kuttner \(2001\)](#). W_{it} may include following control variables: $\Delta \log \text{bid-ask spread}_{it}$ is the log change in the monthly average relative bid-ask spread and $\Delta \log S_{it}$ is the log change in the spot FX rate expressed as the number of foreign currency units per US dollar. Risk denominator and independent variables are measured in units of standard deviation. *10

Table 9: economic sources of currency risk.

Currency-risk robustness tables. The tables compare the carry factor with deeper economic characteristics and alternative currency-risk portfolios.

Table 8
Firm-level outcomes and the FX factor structure.

	Δ log debt _{it,j}	Δ log assets _{it,j}	Δ log PPE _{it,j}	Δ log debt _{it,j}	Δ log assets _{it,j}	Δ log PPE _{it,j}
carry beta _{it}	2.25* [1.74]	0.65 [0.77]	0.93 [0.90]	2.24* [1.69]	0.78 [0.91]	0.79 [0.72]
carry beta _{it} × MPS _{it}	1.54** [2.02]	1.16** [2.39]	1.19*** [7.41]	1.50** [2.29]	1.03** [2.41]	1.05*** [5.83]
Δ log S _{it}	7.66*** [4.76]	-0.21 [0.35]	0.75 [1.37]	7.34*** [4.54]	-0.35 [0.60]	0.74 [1.30]
log assets _{it,t-1}	-8.36*** [7.07]	-6.74*** [11.48]	-3.00*** [5.75]	-8.64*** [7.04]	-6.73*** [11.59]	-2.86*** [5.80]
current asset ratio _{it,t-1}	-0.38 [0.85]	-0.73** [2.39]	3.39*** [8.39]	-0.48 [1.07]	-0.81*** [2.63]	3.34*** [8.06]
Δ log sales _{it}	3.32*** [8.36]	2.69*** [14.42]	2.47*** [7.41]	3.30*** [8.54]	2.67*** [14.42]	2.39*** [8.06]
Δ log leverage _{it}	19.90*** [8.65]	1.52*** [4.42]	2.76*** [4.95]	19.88*** [8.80]	1.55*** [4.67]	2.71*** [5.14]
Country × quarter FE	-3.46*** [3.64]	-1.99*** [4.45]	-1.40*** [3.75]	-4.87*** [3.49]	-2.82*** [4.29]	-1.97*** [3.79]
Industry × quarter FE	No	No	No	No	No	No
Overall R ² in %	48.66	27.61	17.82	49.87	29.89	20.69
Total # of Obs.	50498	50498	50498	50498	50498	50498
Avg. #Time periods	51	51	51	51	51	51
#Firms	987	987	987	987	987	987
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Country × quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry × quarter FE	No	No	No	No	No	No

Note: This table reports results from fixed effects panel regressions of the form $\Delta \log Y_{it,j} = \mu_i + \alpha_i + \gamma_i X_{it} + \phi(X_{it} \times MPS_{it}) + \psi \Delta \log S_{it} + \omega W_{it} + \epsilon_{it,j}$, where $\Delta \log Y_{it,j}$ is the quarterly growth rate (in percentage points, that is, point estimates are multiplied times 100) of a firm-level outcome variable Y (see column header) for firm j in country c borrowing in currency i at time t . We winsorize the dependent variable at the 0.5% level. X_{it} denotes the carry beta, that is based on a rolling window regression of currency returns on the carry factor. MPS_{it} is our monetary policy shock in basis points that we extract from Fed Fund futures rate changes following [Kuttner \(2001\)](#). $\Delta \log S_{it}$ is the log change in the spot FX rate expressed as the number of foreign currency units per US dollar and W_{it} includes the same controls as in Equation 11 in the paper. μ_i , α_i , and γ_i denote firm, quarter times firm, and quarter times industry α fixed effects, respectively. AME in % is the average marginal effect for the point estimate concerning either the interaction of dollar or carry beta and $\Delta \log S_{it}$.

Table 8: debt, assets, and PPE outcomes.

Table 10
Currency flows of funds and a comparison of currency risks.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
MPS _{it}	0.01 [0.26]	0.01 [0.34]	0.02 [0.35]	0.01 [0.11]	-0.01 [0.23]	0.05* [1.76]	0.00 [0.05]
carry beta _{it} × MPS _{it}	0.09*** [2.73]						
1M momentum beta _{it} × MPS _{it}		-0.03** [2.00]					
3M momentum beta _{it} × MPS _{it}			-0.04 [0.91]				
1Y momentum beta _{it} × MPS _{it}				-0.05 [1.28]			
value beta _{it} × MPS _{it}					0.06 [1.15]		
output gap beta _{it} × MPS _{it}						-0.03*** [6.47]	
skewness beta _{it} × MPS _{it}							0.05 [1.50]
Δ log bid-ask spread _{it}	-0.03 [1.34]	-0.03* [1.68]	-0.02 [1.09]	-0.03 [1.32]	-0.03* [1.67]	-0.02 [1.19]	-0.02 [1.50]
Δ log S _{it}	-0.01 [0.38]	-0.02 [0.60]	-0.02 [0.59]	-0.02 [0.84]	-0.02 [0.59]	-0.02 [0.59]	-0.02 [0.70]
Overall R ² in %	19.45	19.57	19.50	21.78	19.15	19.70	20.25
Total # of Obs.	1242	1242	1242	1242	1242	1242	1242
Avg. #Time periods	138	138	138	138	138	138	138
#Currencies	9	9	9	9	9	9	8
Currency FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note: This table reports results from fixed effects panel regressions of the form $OF(S_{it}) = \mu_i + \gamma_i X_{it} + \beta MPS_{it} + \phi(X_{it} \times MPS_{it}) + \alpha W_{it} + \epsilon_{it}$, where $OF(S_{it})$ is the β order flow by funds in \$ in currency pair i in month t . X_{it} denotes the exposure (i.e., betas) to various FX risk factors. For convenience, we only report the β and ϕ coefficients. MPS_{it} is our monetary policy shock in basis points that we extract from Fed Fund futures rate changes following [Kuttner \(2001\)](#). W_{it} may include the following control variables: $\Delta \log \text{bid-ask spread}_{it}$ is the log change in the monthly average relative bid-ask spread and $\Delta \log S_{it}$ is the log change in the spot FX rate expressed as the number of foreign currency units per US dollar. Risk denominator and independent variables are measured in units of standard deviation. *10

Table 10: alternative currency risk factors.

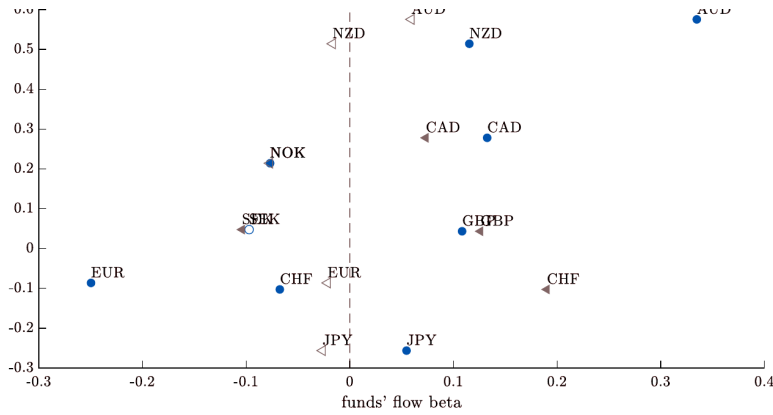


Fig. 6: easing versus tightening. The visual separates fund-flow betas for monetary easing and tightening episodes and shows that the risk-on carry-beta slope is strongest during easing.

6.7 Table 11: Currency flows of funds in response to alternative monetary policy shocks; Table 12: Foreign currency loans in response to alternative monetary policy shocks

Read: Tables 11 and 12 replace the baseline Kuttner shock with alternative high-frequency monetary policy shocks from Kearns et al., Jarociński and Karadi, and Bauer-Swanson.

Detailed interpretation: The main result survives for the target-factor shocks most directly linked to policy-rate surprises. It is weaker for news or central-bank-information components, which is consistent with the idea that the risk-on channel is tied to unexpected easing rather than all FOMC-related information.

How it fits the paper: These tables rule out the possibility that the baseline results are an artifact of one shock measure.

Table 11
Currency flows of funds in response to alternative monetary policy shocks.

	Kuttner (2001)	Kearns et al. (2022)			Jarociński and Karadi (2020)		Bauer and Swanson (2023b)	
	Target	Path	Long-rate	MP	CHI	NS	ORT	
carry beta _{it}	-0.15 [1.10]	-0.13 [1.01]	-0.13 [1.00]	-0.13 [1.04]	-0.14 [1.05]	-0.12 [0.94]	-0.10 [0.99]	-0.10 [0.87]
dollar beta _{it}	0.20 [0.97]	0.17 [0.90]	0.18 [0.91]	0.18 [0.95]	0.18 [0.95]	0.17 [0.94]	0.15 [0.97]	0.15 [0.87]
carry beta _{it} × MPS _{it}	0.17*** [4.18]	0.11*** [2.07]	0.11*** [1.20]	0.01 [0.95]	0.04* [0.40]	0.05 [1.91]	0.05 [1.12]	0.05 [0.79]
dollar beta _{it} × MPS _{it}	-0.36*** [3.68]	-0.12* [1.79]	0.04 [0.76]	0.05 [0.69]	0.05 [0.72]	0.04 [0.59]	-0.01 [0.17]	-0.08 [1.36]
Δ log bid-ask spread _{it}	0.03 [0.92]	0.00 [0.07]	-0.02 [0.76]	-0.02 [0.82]	-0.02 [0.83]	-0.01 [0.75]	-0.01 [0.78]	-0.01 [0.26]
Δ log S _{it}	-0.02 [0.47]	-0.03 [0.62]	-0.04 [0.96]	-0.05 [1.00]	-0.04 [0.94]	-0.05 [1.02]	-0.05 [1.23]	-0.06 [1.53]
Overall R ² in %	31.88	32.41	31.75	31.72	31.70	31.78	32.06	32.78
Total # of Obs.	1242	1233	1233	1233	1233	1233	1224	1161
Avg. #Time periods	138	137	137	137	137	137	136	129
#Currencies	9	9	9	9	9	9	9	9
Currency FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time series FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note: This table reports results from fixed effects panel regressions of the form $OF(S_{it}) = \mu_i + \alpha_i + \gamma X_{it} + \rho(X_{it} \times MPS_{it}) + \sigma W_{it} + \epsilon_{it}$, where $OF(S_{it})$ is the order flow by funds in \$bn in currency pair i in month t . X_{it} denotes either the carry beta_{it} or dollar beta_{it}, that are based on rolling window regressions of currency returns on the carry and dollar factor, respectively. MPS_{it} is a monetary policy shock shown in the column headers. W_{it} includes the following control variables: $\Delta \log \text{bid-ask spread}_{it}$ is the log change in the monthly average relative bid-ask spread and $\Delta \log S_{it}$ is the log change in the spot exchange rate expressed as the number of foreign currency units per unit of US dollar. Both dependent and independent variables are measured in units of standard deviations. The test statistics based on double clustered (by currencies and time) standard errors, allowing for serial correlation up to 3 lags are reported in brackets. Asterisks *, **, and *** denote significance at the 90%, 95%, and 99% confidence levels.

Table 11: alternative shocks for funds.

Robustness to monetary policy shock definitions.

6.8 Table 13: Predicting foreign policy rates with Fed Fund rates; Table 14: Flow betas for euro currency pairs

Read: Table 13 tests whether foreign central banks' policy-rate responses to the Fed explain the risk-on pattern. Table 14 repeats flow-beta estimation for euro-based currency pairs and ECB shocks.

Table 12
Foreign currency loans in response to alternative monetary policy shocks.

	Kuttner (2001)	Kearns et al. (2022)			Jarociński and Karadi (2020)		Bauer and Swanson (2023b)	
	Target	Path	Long-rate	MP	CHI	NS	ORT	
carry beta _{it}	-1.53* [1.94]	-1.50* [1.89]	-1.51* [1.87]	-1.50* [1.89]	-1.51* [1.95]	-1.54* [1.86]	-1.48* [1.81]	-1.48* [1.81]
dollar beta _{it}	0.97* [1.67]	0.94 [1.62]	0.94 [1.60]	0.93 [1.60]	0.93 [1.65]	0.96* [1.65]	0.91 [1.56]	0.95 [1.59]
carry beta _{it} × MPS _{it}	0.21** [2.36]	0.20** [2.24]	0.16*** [4.29]	-0.14*** [2.69]	0.23* [1.83]	0.14 [0.96]	0.22* [1.84]	0.37*** [4.35]
dollar beta _{it} × MPS _{it}	-0.87*** [2.82]	-0.37 [1.16]	0.13*** [4.68]	-0.06*** [2.69]	-0.13 [0.25]	-1.13*** [3.62]	-0.65 [0.94]	0.24 [0.36]
Δ log S _{it}	-0.02 [0.11]	-0.01 [0.03]	-0.03 [0.13]	-0.02 [0.09]	-0.02 [0.11]	-0.06 [0.34]	-0.01 [0.04]	-0.06 [0.35]
Overall R ² in %	62.73	62.69	62.68	62.67	62.69	62.77	62.71	62.77
Total # of Obs.	2727	2727	2727	2727	2727	2727	2727	2727
Avg. #Time periods	303	303	303	303	303	303	303	303
#Currencies	9	9	9	9	9	9	9	9
Currency FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time series FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note: This table reports results from fixed effects panel regressions of the form $\log Loan_{it} = \mu_i + \alpha_i + \gamma X_{it} + \rho(X_{it} \times MPS_{it}) + \sigma W_{it} + \epsilon_{it}$, where $\log Loan_{it}$ is the natural log of the dollar amount lent by global banks headquartered in the US to corporations domiciled abroad in currency i during month t . X_{it} denotes either the carry beta_{it} or dollar beta_{it}, that are based on rolling window regressions of currency returns on the carry and dollar factor, respectively. MPS_{it} is a monetary policy shock shown in the column headers. $\Delta \log S_{it}$ is the log change in the spot exchange rate expressed as the number of foreign currency units per unit of US dollar. The independent variables are measured in units of standard deviations. The test statistics based on double clustered (by currencies and time) standard errors, allowing for first-order serial correlation are reported in brackets. Asterisks *, **, and *** denote significance at the 90%, 95%, and 99% confidence levels. The sample spans from January 1999 to March 2024.

Table 12: alternative shocks for loans.

Detailed interpretation: Table 13 shows that foreign policy-rate responses do not line up with the alternative story that high-risk currencies simply follow lower foreign rates after Fed easing. Table 14 shows that the effect is not a generic response to any large central bank: euro-based pairs do not display the same systematic pattern in response to ECB monetary policy.

How it fits the paper: These checks reinforce that the channel is specifically about US monetary policy interacting with dollar-based currency risk.

Table 13

Predicting foreign policy rates with Fed Fund rates.

	(1)	(2)	(3)	(4)	(5)
ΔFFR_{t-1}	0.22** [2.00]	0.22** [2.26]	0.23 [0.62]		
carry β_{t-1}		-0.08** [2.17]		-0.12* [1.88]	
dollar β_{t-1}			0.02 [0.26]		0.01 [0.11]
carry $\beta_{t-1} \times \Delta FFR_{t-1}$		0.11** [2.37]		0.07 [0.78]	
dollar $\beta_{t-1} \times \Delta FFR_{t-1}$			-0.01 [0.02]		0.03 [0.07]
Overall R^2 in %	5.23	6.90	5.25	43.35	42.76
Total # of Obs.	2745	2745	2745	2745	2745
Avg. #Time periods	305	305	305	305	305
#Currencies	9	9	9	9	9
Currency FE	Yes	Yes	Yes	Yes	Yes
Time series FE	No	No	No	Yes	Yes

Note: This table reports results from monthly fixed effects panel regressions of the form $\Delta y_{i,t} = \mu_i + \alpha_1 + \beta_1 \Delta FFR_{t-1} + \gamma_1 X_{i,t} + \eta_1(X_{i,t} \times \Delta FFR_{t-1}) + \epsilon_{i,t}$, where the dependent variable is the change in the policy rate by foreign central bank i one period ahead and ΔFFR_{t-1} is the change in the Federal funds target rate. We include both country- and time-fixed effects μ_i and α_1 , respectively. $\epsilon_{i,t}$ denotes either the carry β_{t-1} or dollar β_{t-1} that are based on rolling window regressions of currency excess returns on the carry and dollar factor, respectively. Both dependent and independent variables are measured in units of standard deviations. The test statistics based on double clustered (by currencies and time) standard errors, allowing for serial correlation up to 3 lags are reported in brackets. Asterisks *, **, and *** denote significance at the 90%, 95%, and 99% confidence levels. The sample spans from January 1999 to March 2024.

Table 13: foreign policy-rate response.

Alternative macro-policy explanations.

Table 14

Flow betas for euro currency pairs.

	Corporates	Funds	NBFIs	Non-dealer banks	Dealer banks	Carry beta	Euro beta
EURCHF	-0.22*** [4.14]	-0.12 [0.78]	-0.17* [1.89]	-0.07 [1.09]	0.19* [1.78]	-0.17	0.64
EURJPY	0.08 [1.09]	0.18 [1.34]	0.05 [1.06]	-0.22* [1.86]	0.11* [1.69]	-0.03	1.21
EURDKK	0.04 [0.23]	0.06 [0.89]	0.14 [1.52]	-0.16*** [2.79]	0.09* [1.81]	0.00	0.01
EURGBP	-0.09 [1.01]	0.12 [1.63]	-0.07 [0.85]	-0.15* [1.84]	0.11 [1.38]	0.10	0.99
EURSEK	-0.03 [0.42]	-0.07 [0.98]	0.01 [0.07]	0.12 [1.19]	-0.06 [0.66]	0.17	0.59
EURUSD	-0.10 [0.90]	-0.15 [1.30]	0.09 [1.20]	-0.04 [0.74]	0.16* [1.79]	0.18	1.26
EURNOK	0.06 [0.71]	-0.07 [1.12]	0.04 [0.71]	0.03 [0.40]	0.02 [0.35]	0.33	1.10
EURCAD	-0.04 [0.82]	-0.21 [0.94]	0.15*** [2.33]	0.12* [1.91]	0.02 [0.12]	0.45	1.49
EURAUD	0.02 [0.23]	-0.02 [0.21]	-0.12* [1.93]	0.17 [1.37]	-0.14 [0.93]	0.68	1.71

Note: This table reports the β_{ij} regression coefficients from $OF(S_{i,t}) = \alpha_{ij} + \beta_{ij} MPS_t + \epsilon_{i,t}$, where $OF(S_{i,t})$ is the currency flow in €mn in currency pair i customer group j in month t and MPS_t is our monetary policy shock measure based on Jarocinski and Karadi (2020). Both dependent and independent variables are measured in units of standard deviations. Currency pairs are sorted by carry betas in ascending order. By market clearing, the column labeled “Dealer banks” is equal to the sum of the first four columns times minus one. The last two columns report the average carry and euro beta that we compute based on rolling window regressions. For both the carry and euro factor (average exchange rate change against the euro) we are only using the nine euro-currency pairs displayed in the first column. Asterisks *, **, and *** denote significance at the 90%, 95%, and 99% confidence levels. The numbers inside the brackets are the corresponding test statistics based on robust standard errors (Newey and West, 1987) correcting for heteroskedasticity

Table 14: euro-pair placebo.

6.9 Box I and Fig. A.1: Equilibrium following a 25 bps decline in the Fed Funds rate

Read: Box I reports the calibrated expected returns and covariance matrix used in the model. Fig. A.1 shows how a 25-basis-point decline in the Fed Funds rate changes intermediary currency demand, carry betas, and equilibrium capital rental rates.

Detailed interpretation: The model predicts that lowering the US short rate increases intermediary risk-taking capacity. Currencies with higher expected excess returns and carry betas attract more intermediation capital. The bottom-right panel shows how changes in capital rental rates translate financial reallocation into real investment incentives.

How it fits the paper: The appendix visual provides a theoretical map from the empirical risk-on facts to firm-level real effects.

$$\mathbb{E}[\mathbf{K}'_A] \times 1200 = \begin{bmatrix} 1.760 \\ 0.536 \\ -0.062 \\ -1.040 \\ -0.744 \\ -3.354 \\ -0.708 \\ 2.468 \\ -1.664 \end{bmatrix} \begin{pmatrix} \text{AUD} \\ \text{CAD} \\ \text{CHF} \\ \text{EUR} \\ \text{GBP} \\ \text{JPY} \\ \text{NOK} \\ \text{NZD} \\ \text{SEK} \end{pmatrix}$$

$$\Sigma \times 1200 = \begin{bmatrix} 0.924 & 0.448 & 0.365 & 0.440 & 0.363 & 0.125 & 0.612 & 0.765 & 0.596 \\ 0.448 & 0.397 & 0.176 & 0.231 & 0.221 & 0.041 & 0.356 & 0.363 & 0.322 \\ 0.365 & 0.176 & 0.609 & 0.496 & 0.330 & 0.265 & 0.469 & 0.392 & 0.490 \\ 0.440 & 0.231 & 0.496 & 0.588 & 0.379 & 0.206 & 0.532 & 0.436 & 0.573 \\ 0.363 & 0.221 & 0.330 & 0.379 & 0.526 & 0.108 & 0.425 & 0.382 & 0.422 \\ 0.125 & 0.041 & 0.265 & 0.206 & 0.108 & 0.621 & 0.173 & 0.125 & 0.203 \\ 0.612 & 0.356 & 0.469 & 0.532 & 0.425 & 0.173 & 0.840 & 0.542 & 0.642 \\ 0.765 & 0.363 & 0.392 & 0.436 & 0.382 & 0.125 & 0.542 & 0.945 & 0.567 \\ 0.596 & 0.322 & 0.490 & 0.573 & 0.422 & 0.203 & 0.642 & 0.567 & 0.767 \end{bmatrix}$$

Box I.

Box I: calibrated inputs.

Theoretical mechanism behind the empirical channel.

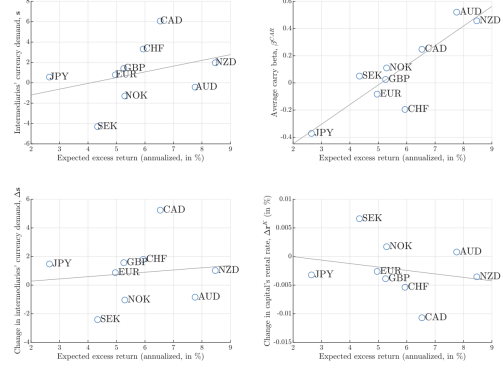


Fig. A.1: Equilibrium following a 25 bps decline in the Fed Funds rate. Note: All x-axes are the model's pre-shock equilibrium excess returns on capital. The top-left figure plots intermediaries' capital supply across countries, s . The top-right figure scatters the average carry betas in the data. The bottom two panels illustrate the comparative statics in response to subtracting 25 basis points from the domestic interest rate r . The bottom-left panel shows the change in demand $s' - s$ and the bottom-right panel shows the change in equilibrium excess returns.

Fig. A.1: model equilibrium after easing.

7 Short summary

- The paper studies the international transmission of US monetary policy through the factor structure of exchange rates.
- The main risk measures are carry betas, dollar betas, and related currency factor exposures.
- High-frequency FOMC surprises identify unexpected US monetary policy shocks.
- CLS data show that investment funds buy high-risk currencies and sell safe currencies after US monetary easing.
- The flow response persists for several months.
- DealScan data show that global US banks tilt foreign currency loan origination toward high-carry-beta currencies after easing.
- Firm-level data show that firms operating in high-risk currencies borrow more, lever up, expand assets, and increase PPE investment.
- The paper controls for exchange-rate movements, country-time shocks, industry-time shocks, and firm characteristics.
- Robustness checks show that the results are not driven by alternative currency factors, foreign central bank policy responses, ECB shocks, or one particular monetary policy shock measure.
- The central message is that US monetary policy is transmitted abroad through cross-currency risk reallocation, not only through aggregate dollar liquidity or interest-rate differentials.

8 Conclusion

8.1 Core conclusion

Loualiche, Pecora, Somogyi, and Ward show that US monetary policy affects the world through the exchange-rate factor structure. Monetary easing induces investors and banks to move up the currency-risk spectrum, allocating more capital to high-carry and high-risk currencies.

8.2 Economic intuition

The intuition is that US easing raises the risk-taking capacity or lowers the effective risk aversion of global intermediaries. When this happens, risky currencies become more attractive. Investment funds respond by buying high-carry-beta currencies, and banks respond by extending relatively more credit in these currencies. Because firms borrow and operate in different currencies, this financial reallocation changes firm balance sheets and investment.

8.3 Incremental contribution revisited

The paper's contribution is to make the currency factor structure operational for international monetary transmission. It shows that currency betas are not just asset-pricing objects; they predict real capital allocation after US policy shocks. The paper also contributes by documenting the mechanism across three linked markets: spot currency flows, syndicated loans, and firm real outcomes.

8.4 Why the result matters

The result matters because it changes how we think about global spillovers. A US easing is not simply a uniform relaxation of global financial conditions. It changes the composition of global risk-taking. The same policy shock can expand financing for firms tied to risky currencies while having weaker or even opposite effects for firms tied to safe currencies.

8.5 Limitations and caveats

The analysis is strongest for countries and firms represented in CLS, DealScan, and Compustat. The paper observes institutional order flows and syndicated lending, but not every form of currency trading or bank credit. It also cannot fully separate all possible risk-taking motives from balance-sheet constraints, although the robustness tests strongly support the currency-risk interpretation.

8.6 Takeaway

The main takeaway is that the exchange-rate factor structure is a quantitatively important channel of monetary policy transmission. Systematic currency risk shapes who receives capital after US monetary easing, and this allocation has measurable effects on firm leverage and investment abroad.